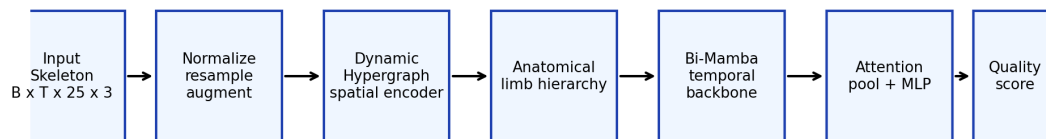


Final Strategic Decision and Implementation Plan

Repo implementation -> your plan -> MAMBA/JAMBA/TTT/xLSTM comparison -> one core topic -> ablation strategy -> execution roadmap

Final decision: choose **Dynamic Hypergraph Hierarchical Mamba Network for Automated Assessment of Physical Rehabilitation Exercises** as the main research topic. Use the original GitHub repo as Baseline 0, use your H-ST-Mamba/DHGNN/pretraining plan as the target framework, and use xLSTM, Jamba-style, and TTT as comparative temporal backbones.



Core topic: Dynamic Hypergraph Hierarchical Mamba Network for Automated Assessment of Physical Rehabilitation Exercises

Baselines: repo Transformer, clean Transformer, xLSTM, Jamba-style, TTT. Main model: Bi-Mamba + DHGNN + hierarchy + physiology-aware pretraining.

1. The core topic we should commit to

Core research title: Dynamic Hypergraph Hierarchical Mamba Network with Physiology-Informed Contrastive Pretraining for Automated Assessment of Physical Rehabilitation Exercises.

This is the strongest topic because it is specific enough to implement, broad enough to publish, and directly evolves from the repo and your uploaded plan. The original repo gives a CurveNet + Transformer baseline. Your uploaded plan argues that the current Transformer-style approach misses dynamic joint importance, anatomical hierarchy, long-sequence efficiency, and pretraining for small rehabilitation datasets. The best decision is to keep these ideas, but make **Mamba** the main temporal backbone rather than trying to make every new architecture the main method.

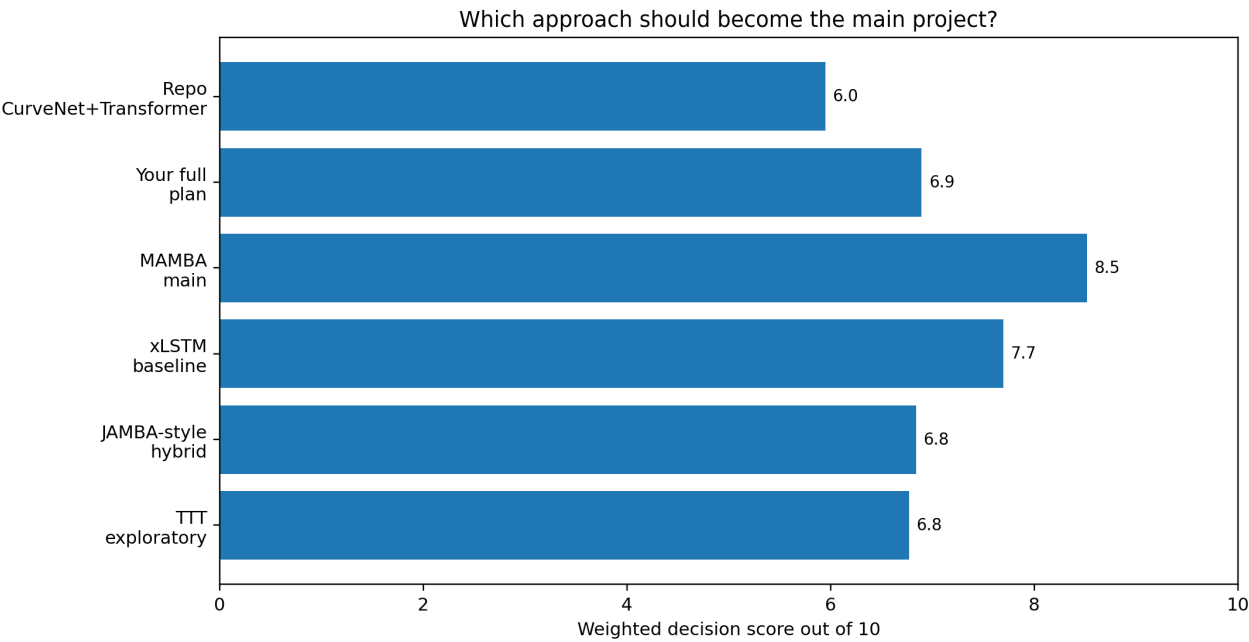
Why this should be the main goal

- It has a clean paper story: baseline Transformer rehab -> dynamic hypergraph -> hierarchy -> Mamba temporal modeling -> physiology-aware pretraining.
- It fits your resource constraint better than a Jamba-style or TTT-first strategy.
- It allows strong ablations: remove hypergraph, remove hierarchy, replace Mamba with xLSTM/Jamba/TTT, remove pretraining, remove anatomical loss.
- It is not just a model swap. It connects temporal modeling with rehabilitation biomechanics and remote assessment.

Do not frame the paper as: Mamba beats Transformer always. **Frame it as:** an anatomical dynamic hypergraph + hierarchical Mamba design improves rehabilitation assessment under fair subject-wise evaluation.

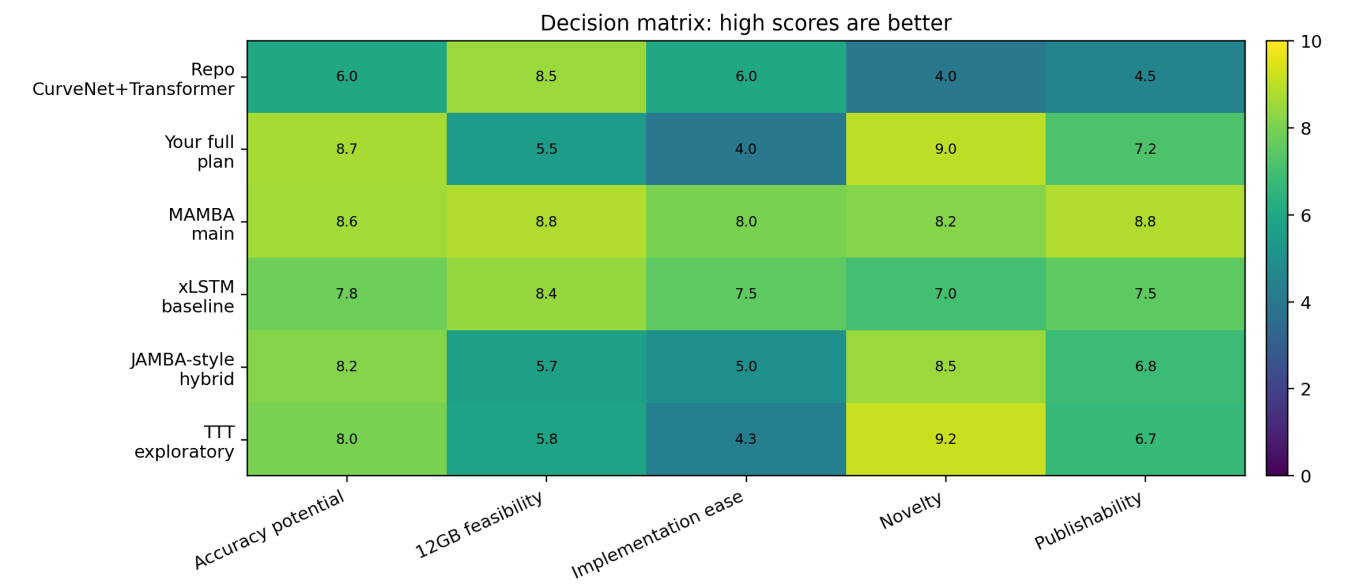
2. What we are comparing

Candidate	Role in project	Use in final paper
Repo implementation: CurveNet+Transformer	Baseline to reproduce and repair as much as possible.	Stable baseline to beat.
Your uploaded plan: H-ST-Mamba + DeepMN Contrastive	Strategic blueprint containing publishable directions.	Converted into a staged, testable implementation.
MAMBA / H-ST-Mamba	Main model.	Primary contribution.
xLSTM	Stable low-risk temporal baseline.	Strong comparison against Mamba.
JAMBA-style	Hybrid attention + Mamba + MoE comparison.	Optional high-capacity ablation if time/GPU allow.
TTT	High-novelty adaptive temporal comparison.	Exploratory extension, not first target.



3. Final decision matrix

The decision scores below are strategic research scores, not final experimental results. They combine expected accuracy potential, 12GB VRAM feasibility, implementation difficulty, novelty, and publication value. Actual accuracy must be measured after training on UI-PRMD/KIMORE under subject-independent splits.

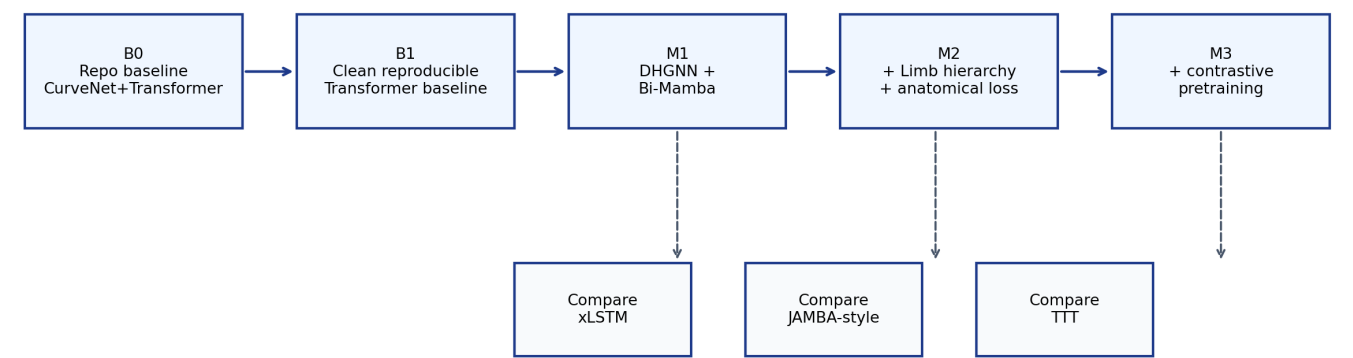


Interpretation

- **MAMBA wins** because it is the best balance of novelty, feasibility, and direct alignment with your plan.
- **xLSTM is second** because it is stable and resource-friendly, making it a strong baseline or backup main method.
- **JAMBA-style is third** because it can test attention + Mamba + expert routing, but it adds complexity and overfitting risk.
- **TTT is fourth** because it is very novel but difficult to implement rigorously and slow for a first publishable result.
- **The repo is not the final goal**, but it is essential as the first reproducible baseline.

4. Strategic evolution path

Recommended evolution path from repo implementation to publishable model



Strategic rule: do not combine everything at first. Prove each component with ablation, then build the final model.

The path should be sequential. Do not implement the full proposed model immediately. First reproduce the repo baseline, then build a clean baseline, then add components one by one. The main model becomes M3: DHGNN + limb hierarchy + Bi-Mamba + anatomical loss + optional contrastive pretraining.

5. Implementation blueprint

The implementation should use one shared data pipeline and one shared spatial/anatomical pipeline. The temporal backbone should be the replaceable module. This makes the comparison scientifically fair.

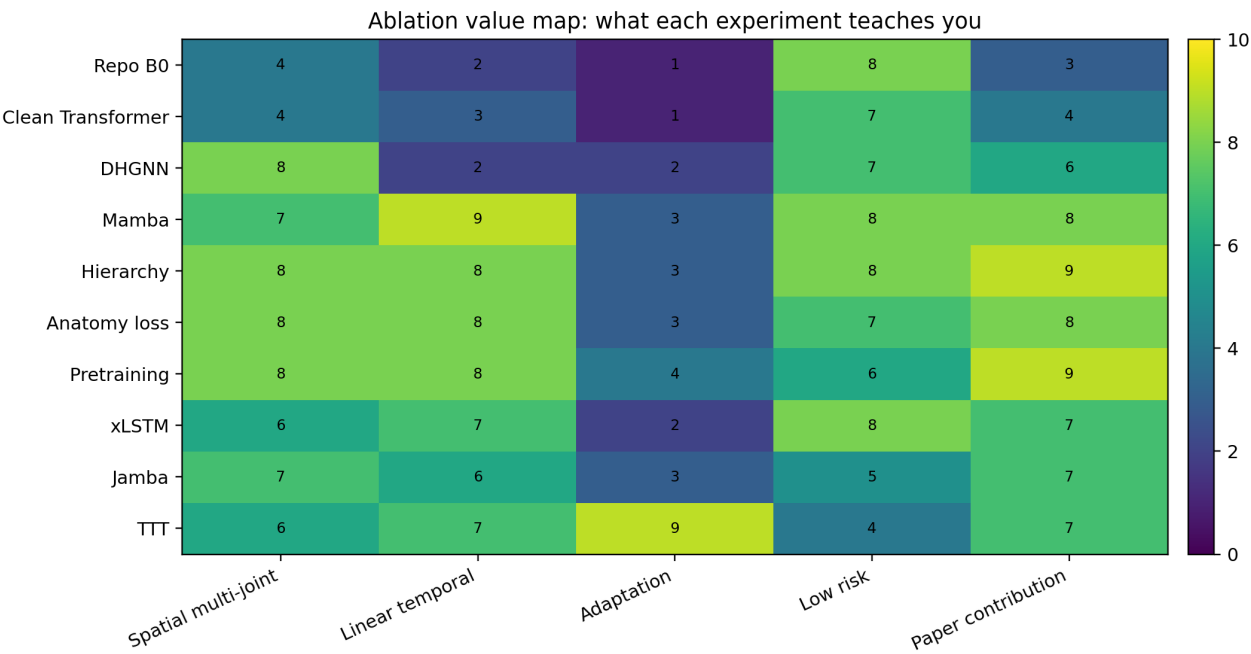
Module	Implementation choice	Why
Data format	NPZ arrays: X=(N,T,J,C), y=(N,), groups=(N,)	Easy to reuse for all methods.
Preprocessing	root-center, scale-normalize, temporal resample to T=200	Reduces subject/camera variation.
Spatial encoder	Dynamic soft hypergraph convolution	Captures multi-joint kinetic chains.
Hierarchy	Pool joints into torso, arms, legs	Matches rehabilitation biomechanics.
Temporal backbone	Bi-Mamba as main; xLSTM/Jamba/TTT as swappable alternatives.	Facilitates comparison.
Head	Temporal attention pool + MLP score regressor	Outputs movement quality score.
Loss	MSE + optional anatomical smoothness/bone-length loss	Supports clinical quality regression.
Evaluation	MAE, RMSE, R2, Pearson; optional threshold accuracy	Avoids relying on only accuracy.

Minimal model pseudo-code

```
class FinalRehabModel(nn.Module):
    def __init__(self, temporal_backbone="mamba"):
        self.spatial = DynamicHypergraphConv(in_dim=3, hidden_dim=96)
        self.hierarchy = LimbPool(in_dim=96, out_dim=128)
        self.temporal = build_temporal_backbone(temporal_backbone, dim=128)
        self.head = TemporalAttentionHead(dim=256)

    def forward(self, x):
        # x: (B, T, 25, 3)
        x = normalize_and_augment(x)
        x = self.spatial(x)           # (B, T, 25, 96)
        x = self.hierarchy(x)         # (B, T, 5, 128)
        x = self.temporal(x)          # (B, T, 5, 256)
        score = self.head(x)          # (B, 1)
        return score
```

6. Ablation study: the experiment plan that proves the paper



Ablation table to execute

ID	Model / experiment	Question answered	Decision rule
A0	Original repo CurveNet + Transformer	What is the starting baseline?	Must reproduce before claiming improvement.
A1	Clean Transformer baseline with same data pipeline	Is there improvement from better code/data pipeline?	Use as controlled baseline.
A2	DHGNN + Transformer	Does dynamic hypergraph help without Mamba?	Keep DHGNN if it improves MAE/accuracy and robustness.

ID	Model / experiment	Question answered	Decision rule
A3	DHGNN + Bi-Mamba, no hierarchy	Does Mamba help temporal modeling?	Keep Mamba if it improves or reduces VRAM/time.
A4	DHGNN + hierarchy + Bi-Mamba	Does anatomical hierarchy help?	Main model if A4 > A3.
A5	A4 + anatomical loss	Does physiology constraint stabilize training?	Keep if robustness improves without hurting clean accuracy.
A6	A5 + contrastive pretraining	Does NTU/pretraining solve small-data overfitting?	Keep if cross-dataset and 5-fold mean improve.
A7	Replace Mamba with xLSTM	Is xLSTM a stronger stable baseline?	Report even if xLSTM wins; it becomes backup.
A8	Replace Mamba with Jamba-style hybrid	Does attention + SSM + MoE help?	Only keep if it clearly beats Mamba under same VRAM.
A9	Replace Mamba with TTT	Does adaptive sequence memory improve patient generalization?	Use as exploratory extension.

7. Final paper structure

Section	Content
Introduction	Remote rehabilitation needs automated quality assessment; current Transformer/graph models miss dynamic anatomical hierarchy a
Related work	Repo Transformer rehab, ST-GCN/STGA-style skeleton models, Mamba/SSM, xLSTM, Jamba-style hybrid, TTT, contrastive pretrain
Method	Dynamic hypergraph spatial encoder, limb hierarchy, Bi-Mamba temporal backbone, physiology-informed pretraining.
Experiments	UI-PRMD and KIMORE; subject-wise 5-fold CV; cross-dataset generalization; noise robustness.
Ablation	A0-A9 table above. This is the proof section.
Discussion	Why Mamba helps or does not help; limitations; deployment feasibility.
Conclusion	A reproducible dynamic hypergraph hierarchical Mamba framework for rehabilitation assessment.

8. Exact evaluation metrics and plots

- **Regression metrics:** MAE, RMSE, R2, Pearson correlation for movement quality score.
- **Classification metrics:** accuracy, balanced accuracy, macro-F1, confusion matrix if labels are correct/incorrect classes.
- **Robustness:** performance under Gaussian pose noise sigma 0.01, 0.03, 0.05 and frame drop 10%, 20%.
- **Efficiency:** parameters, training time per epoch, inference FPS, max VRAM used.
- **Visualizations:** training curve, prediction-vs-ground-truth scatter, temporal attention heatmap, hyperedge/joint-group importance, failure cases.

Core result tables you must create

Table/Figure	What it should show
Main benchmark table	A0-A9 models against UI-PRMD and KIMORE mean +/- std.
Ablation bar chart	Incremental contribution of DHGNN, hierarchy, Mamba, anatomical loss, pretraining.
Robustness curve	Performance drop as pose noise increases.
Resource scatter plot	VRAM vs MAE or FPS vs MAE.
Prediction scatter	Ground truth score vs predicted score.
Interpretability figure	Important frames and important joint groups for each exercise.

9. 14-week execution plan



Weeks	Action	Output
1-2	Repo reproduction and dataset organization	Baseline repo notes, clean folder, processed NPZ files, smoke tests.
3-4	Clean controlled Transformer baseline	A1 result, clean training/eval scripts, first metric table.
5-6	Build main DHGNN + Bi-Mamba model	A2-A4 results and first paper-worthy model.

Weeks	Action	Output
7-9	Ablations: anatomy loss, xLSTM, Jamba, TTT	A5, A7, A8, A9 comparison.
10-11	Contrastive pretraining and fine-tuning	A6 results, cross-dataset transfer tests.
12-13	Robustness and efficiency tests	Noise curve, resource table, inference time.
14	Paper/report writing	Final manuscript, code README, reproducibility appendix.

10. What to pick now: final decision plan

Pick MAMBA / H-ST-Mamba as the main model. Build your research around Dynamic Hypergraph Hierarchical Mamba. Do not pick Jamba, TTT, or xLSTM as the main model. Use them as comparison approaches. Do not pick the GitHub repo as the final project; use it as the baseline.

Choice	Decision	Reason
Main paper model	DHGNN + Hierarchy + Bi-Mamba + physical brain model	Best balance of novelty, feasibility, and resource fit.
Primary baseline	GitHub repo CurveNet + Transformer	Directly related to your starting project.
Strong backup baseline	xLSTM	Stable and easy to train; good check against Mamba.
Optional advanced baseline	Jamba-style	Tests hybrid attention/SSM/MoE, but resource-heavy.
Experimental novelty baseline	TTT	Interesting adaptation story, but high-risk for first paper.

Decision rule during experiments

- If A4 beats A1/A0 and is more efficient, the paper is already viable.
- If A6 pretraining improves cross-dataset or robustness, it becomes the strongest contribution.
- If xLSTM beats Mamba, keep the topic but rename it around dynamic hypergraph hierarchical temporal modeling and report Mamba honestly as a comparison.
- If Jamba/TTT fail, still publish the failure as part of the ablation because it proves your main model is not just a random architecture choice.

11. Implementation commands

Use the code package already generated for the temporal approaches. The main commands are:

```
# 1. Install
unzip Rehabilitation_Temporal_Approaches_Code.zip
cd temporal_approaches_output/code
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt

# 2. Smoke tests on synthetic data
python train.py --approach mamba --synthetic --synthetic-n 16 --epochs 1 --batch-size 4
--target-len 20
python train.py --approach xlstm --synthetic --synthetic-n 16 --epochs 1 --batch-size 4
--target-len 20
python train.py --approach jamba --synthetic --synthetic-n 16 --epochs 1 --batch-size 2
--target-len 20
python train.py --approach ttt --synthetic --synthetic-n 16 --epochs 1 --batch-size 2 --target-
len 20

# 3. Real data layout
data/processed/X.npy      # (N,T,25,3)
data/processed/y.npy      # (N,)
data/processed/groups.npy # subject IDs for subject-wise splitting

# 4. Train main model
python train.py --approach mamba --data data/processed --epochs 120 --batch-size 8 --target-len
120

# 5. Train comparisons
python train.py --approach xlstm --data data/processed --epochs 120 --batch-size 8 --target-len
120
python train.py --approach jamba --data data/processed --epochs 120 --batch-size 4 --target-len
120
python train.py --approach ttt --data data/processed --epochs 100 --batch-size 4 --target-len
100

# 6. Visualize results
python visualize_results.py --run runs/mamba
python visualize_results.py --run runs/xlstm
python visualize_results.py --run runs/jamba
python visualize_results.py --run runs/ttt
```

12. Risks and how to avoid failure

Risk	Why it can ruin the project	Mitigation
Subject leakage	Same subject in train/test can inflate results.	Use GroupKFold by subject and publish split IDs.

Risk	Why it can ruin the project	Mitigation
Too many models at once	You will not know what caused success/failure.	Ablate in A0-A9 order.
Overclaiming expected accuracy	Reviewers reject unsupported claims.	Call numbers hypotheses until real experiments are done.
Wrong joint mapping	Architecture becomes meaningless if limbs are wrong.	Plot skeletons and verify every joint index.
mamba-ssm install problem	Can block progress early.	Use PyTorch fallback first; install official kernel later.
Small dataset overfitting	Accuracy may collapse on new subjects.	Use augmentation, pretraining, early stopping, and cross-dataset tests.

13. References and evidence used

The uploaded plan states the main motivation: Transformer Rehab uses CurveNet + Transformer, identifies fixed graph/anatomical/temporal limitations, proposes H-ST-Mamba, DHGNN, and pretraining, and lists the expected ablation roadmap and expected final results on pages 1-25. The chart on page 9 gives the expected H-ST-Mamba improvements, page 18 gives pretraining improvement expectations, page 21 gives final proposed results, and page 24 lists the ablation study design.

- Uploaded plan: Technical Deep Learning Research: Beating Transformer Rehabilitation SOTA, pages 1-25.
- King-Rafat Transformer_Rehabilitation GitHub repository: baseline repo implementation.
- Mamba: Linear-Time Sequence Modeling with Selective State Spaces, arXiv:2312.00752.
- Jamba: A Hybrid Transformer-Mamba Language Model, arXiv:2403.19887.
- Learning to Learn at Test Time / TTT Layers, arXiv:2407.04620.
- xLSTM: Extended Long Short-Term Memory, arXiv:2405.04517.